



Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-19



Biofuels

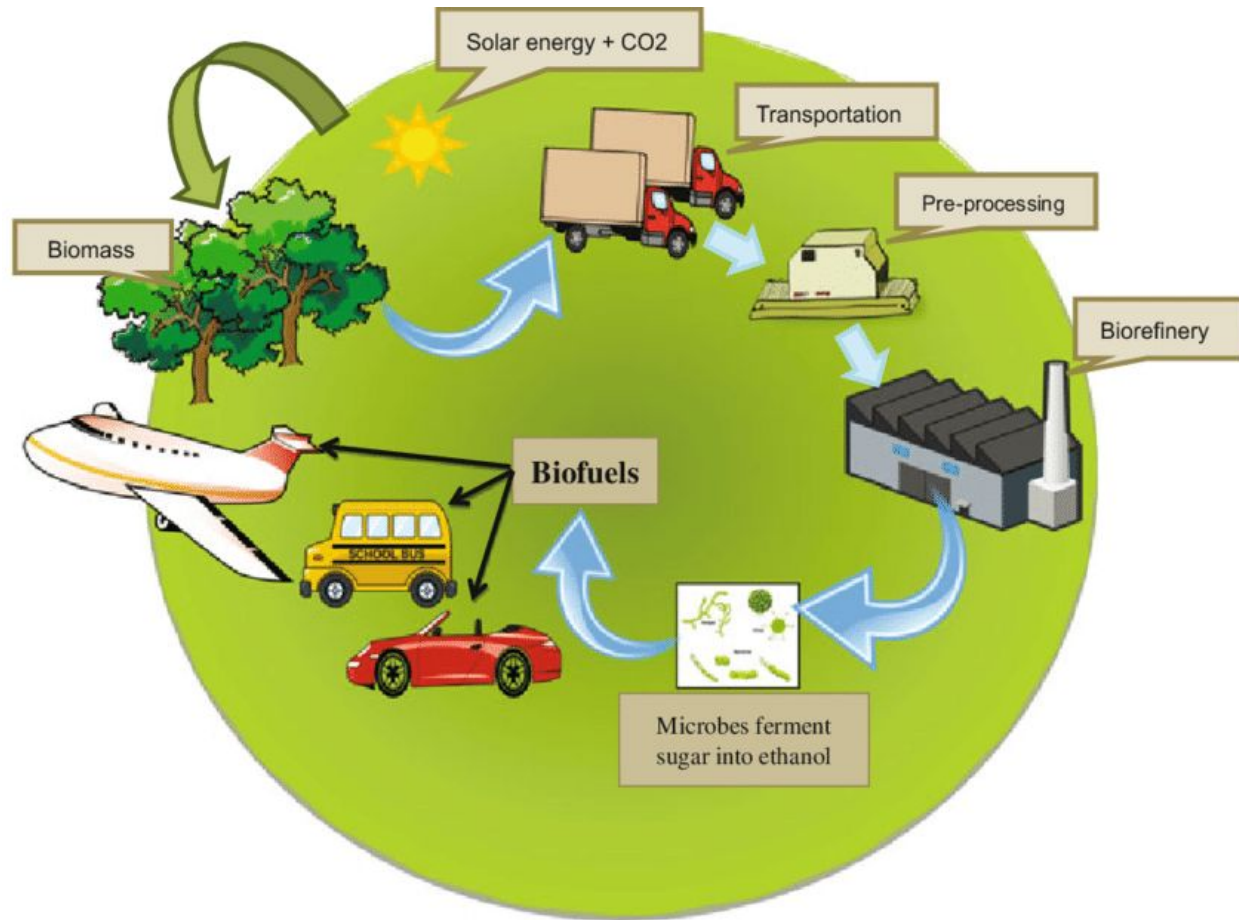
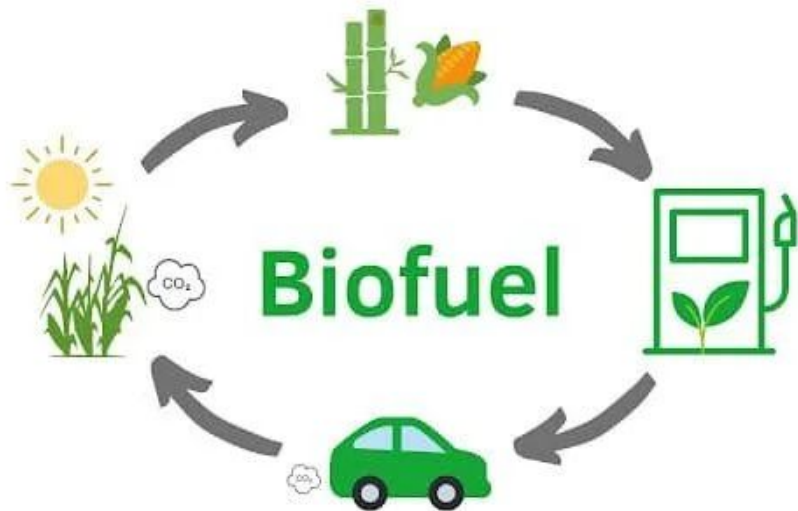
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Biofuels

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“Biofuel is an inexhaustible, biodegradable fuel manufactured from Biomass.”

Biofuel is considered pure and the easiest available fuels on planet earth. Biofuels are obtained from biomass like wood and straw, which are released by direct combustion of dry matter and convert into a gaseous and liquid fuel. Other sources include organic matter like sludge, sewage and vegetable oils matter, which can be converted into biofuels by a wet process like digestion and fermentation.



Types of Biofuels

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Biofuel is generally available in all regions of the world, which mainly include fuels like:

- Biodiesel
- Bioethanol
- Bio methanol

The two most common types of biofuels in use today are bioethanol and biodiesel. Both of which represent the first generation of biofuel technology.



Types of Biofuels

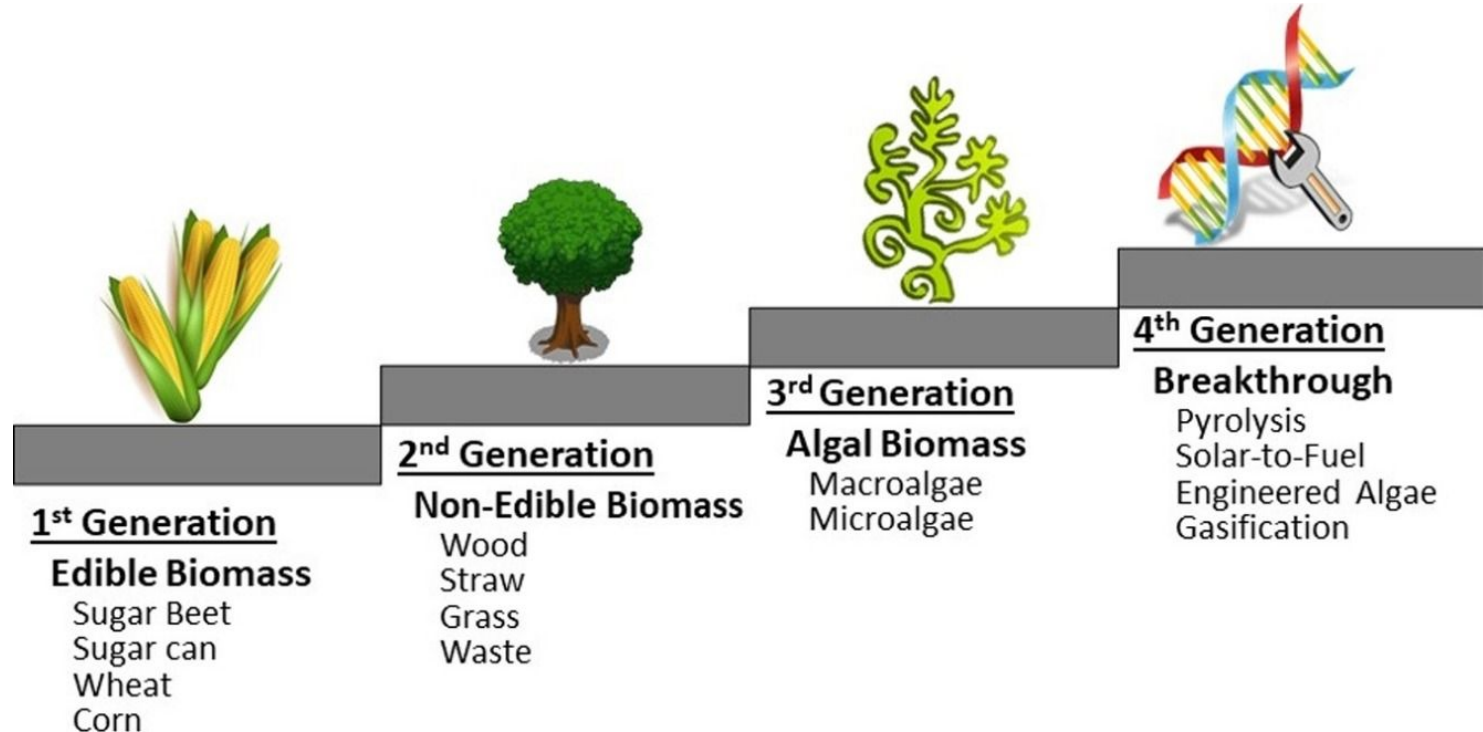


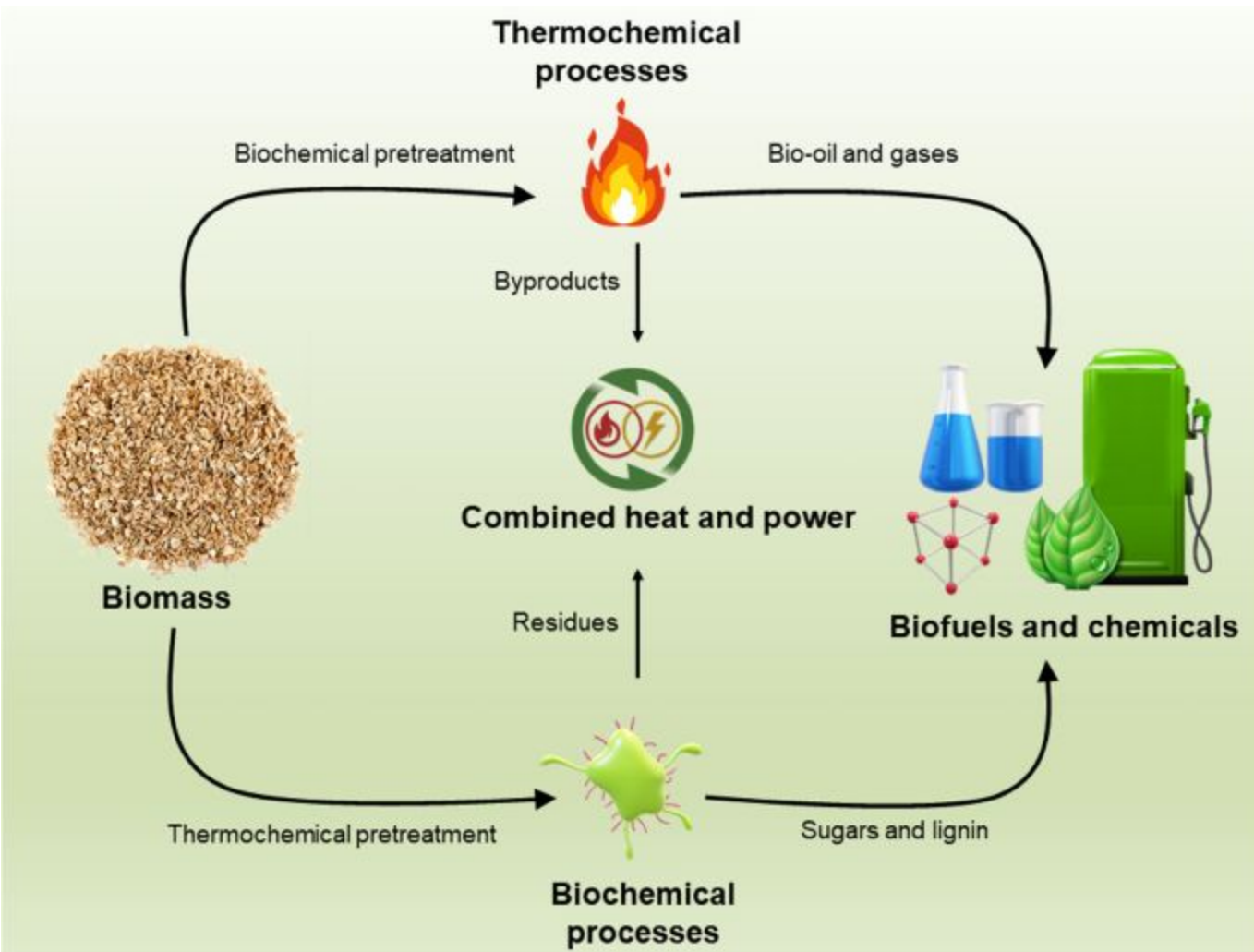
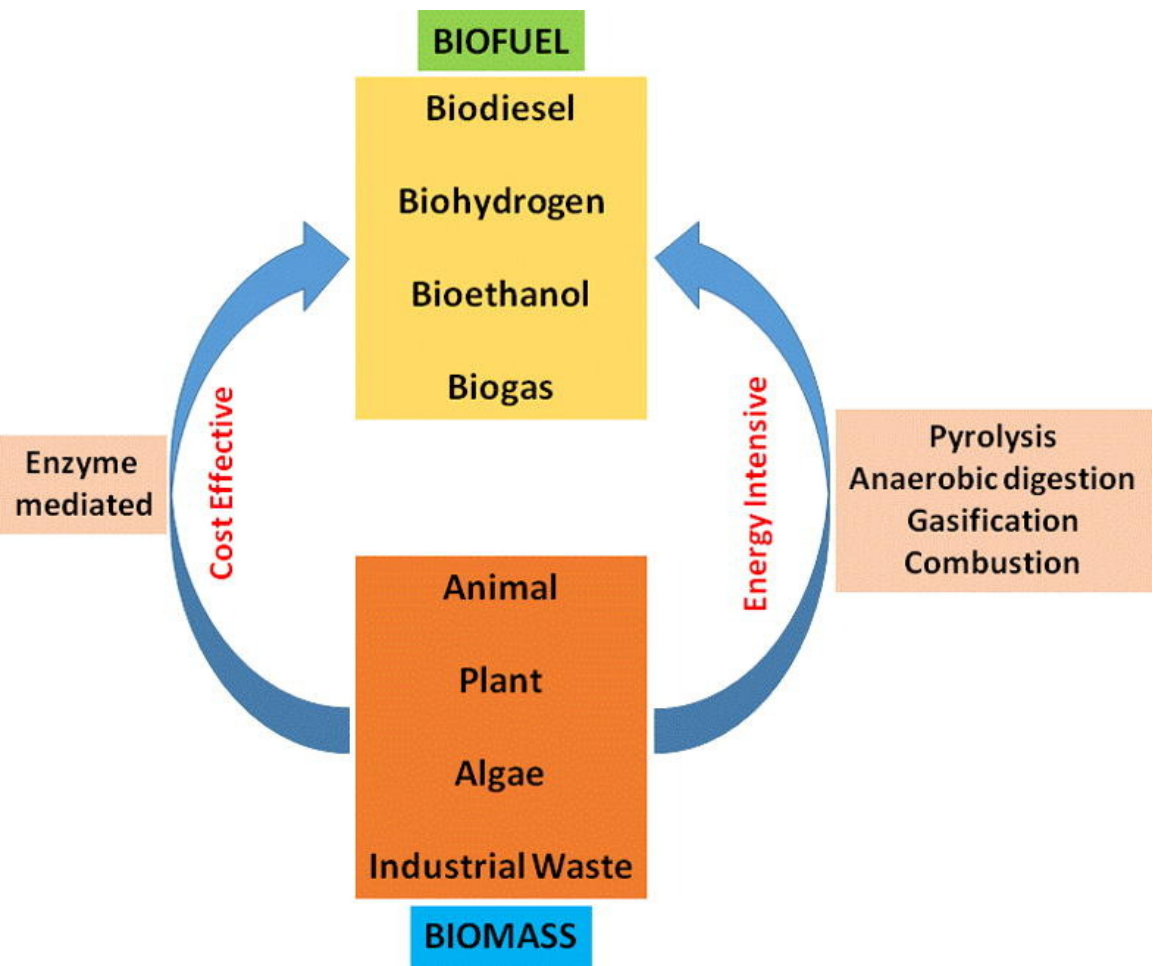
Wood, straw and domestic refuse are used as a source of heat and energy. Biofuels can also be produced from animal fats, plant wastes and other organic wastes produced from living organisms. This process is divided into three categories:

First-generation: Biofuels are primarily produced from edible sugar, starch and other food crops are grown on arable land.

Second generation: Biofuels are produced from non-edible plant materials, such as plant dry matter or woody biomass, or agricultural residues and wastes.

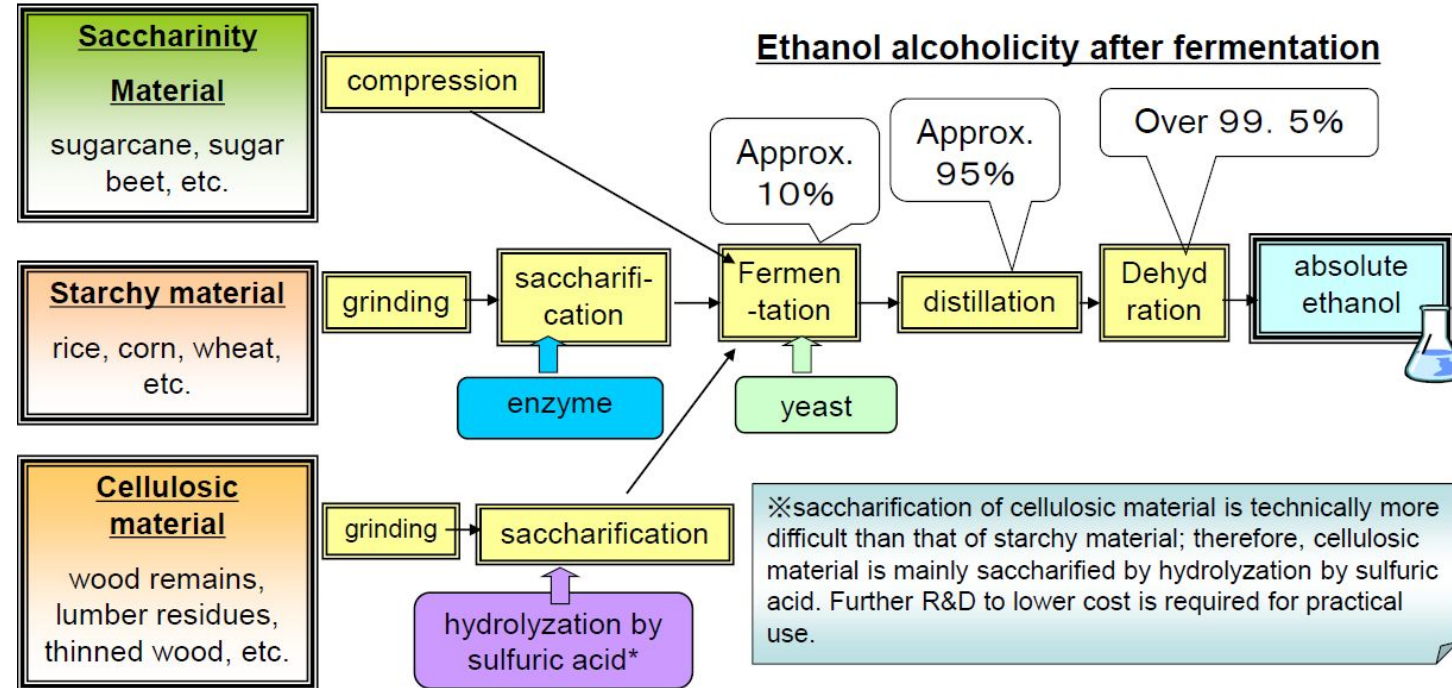
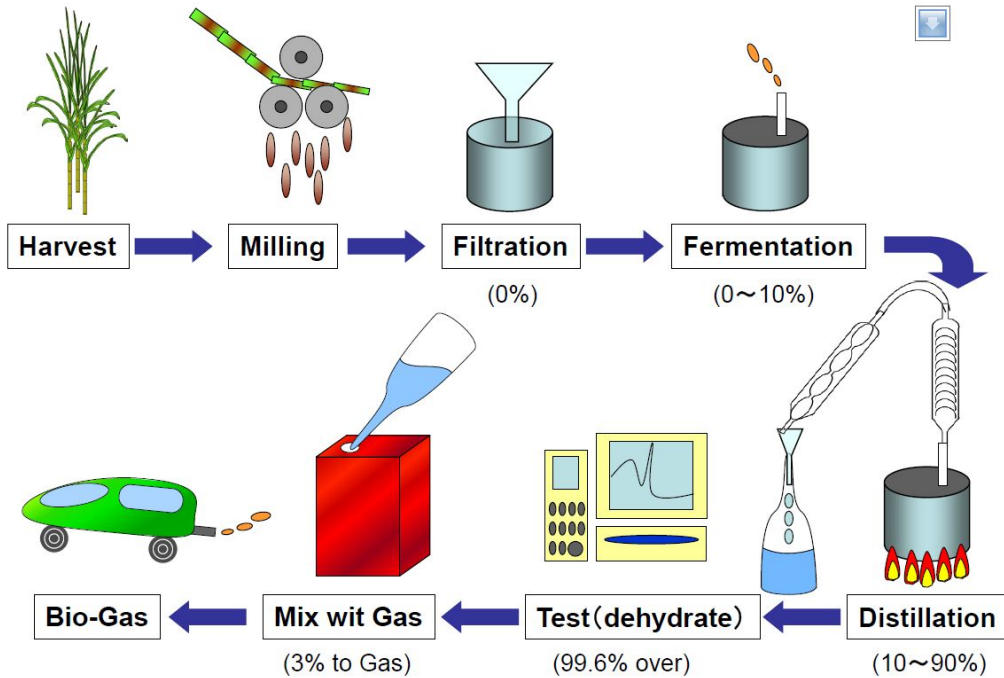
Third generation: Biofuels are primarily produced from algae and microbes. Algal fuels have high yields.

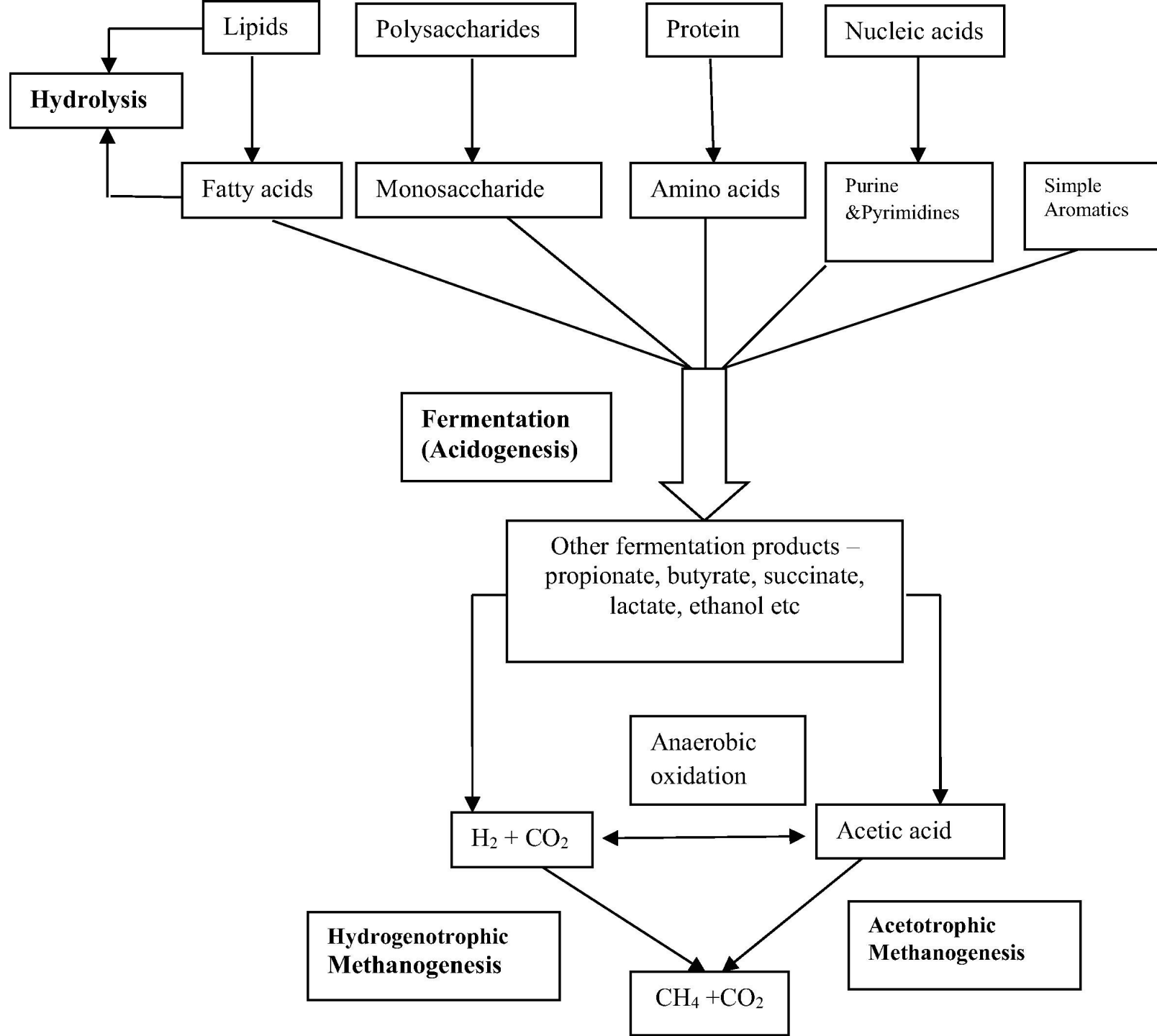




Steps to obtain Biofuels

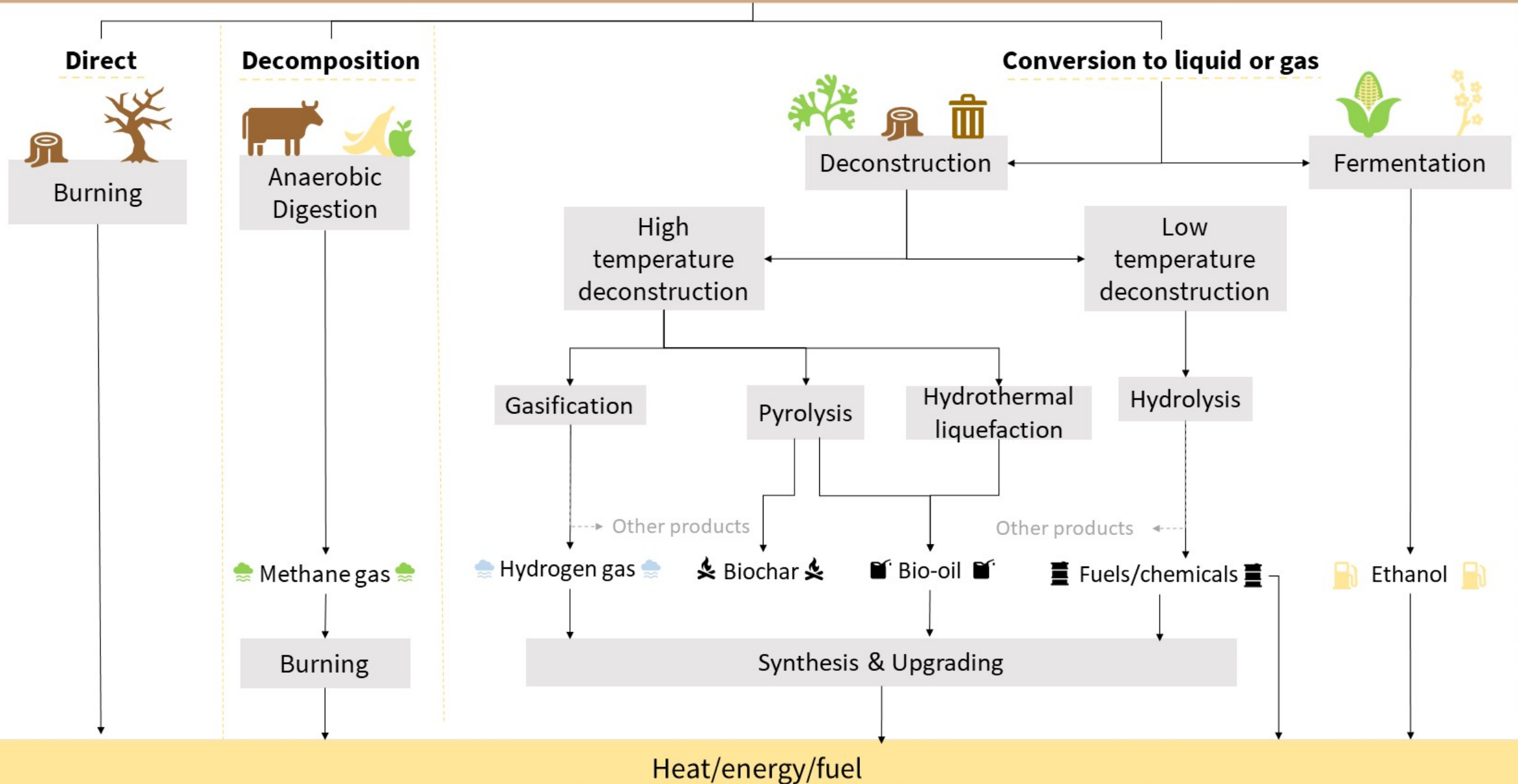
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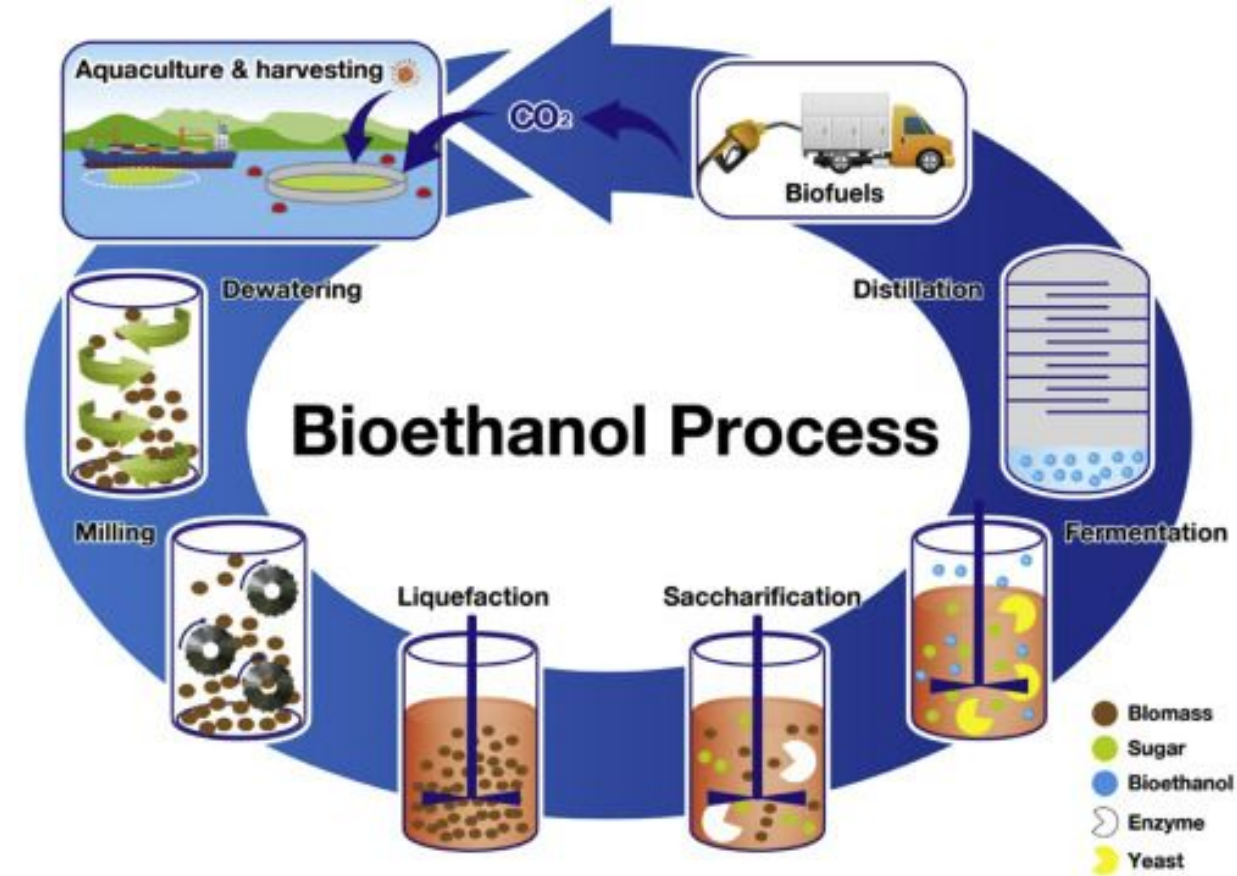
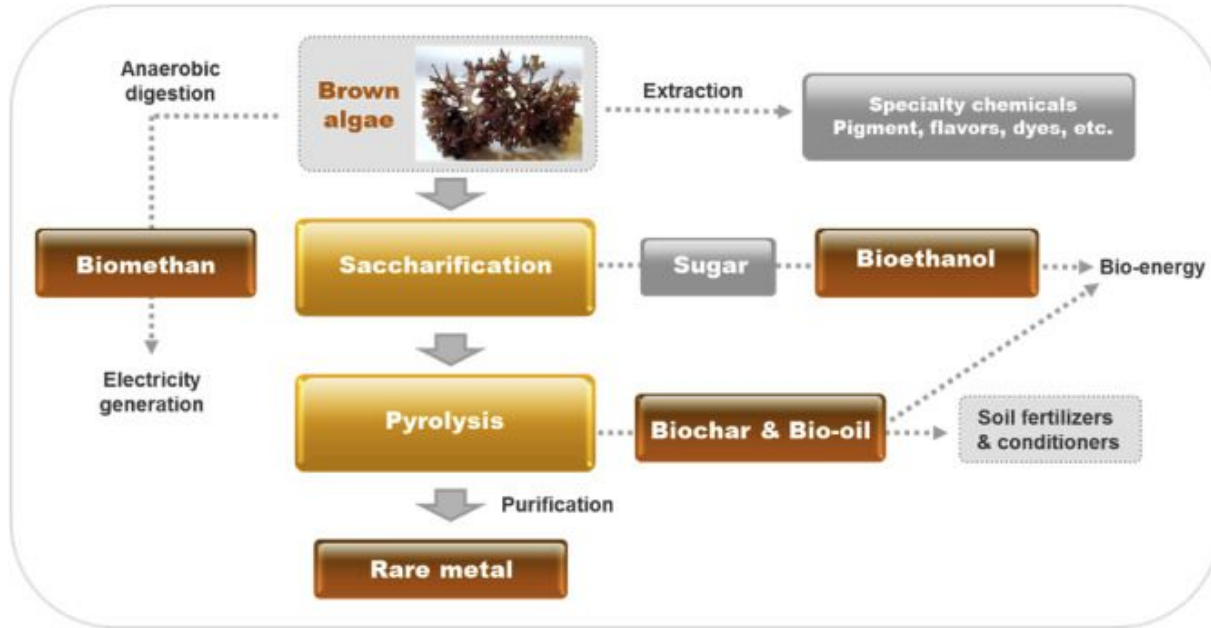


Biomass to Biofuel Processes

Biomass (wood, agricultural waste, animal waste, municipal waste, oilseed crops, algae, etc.)

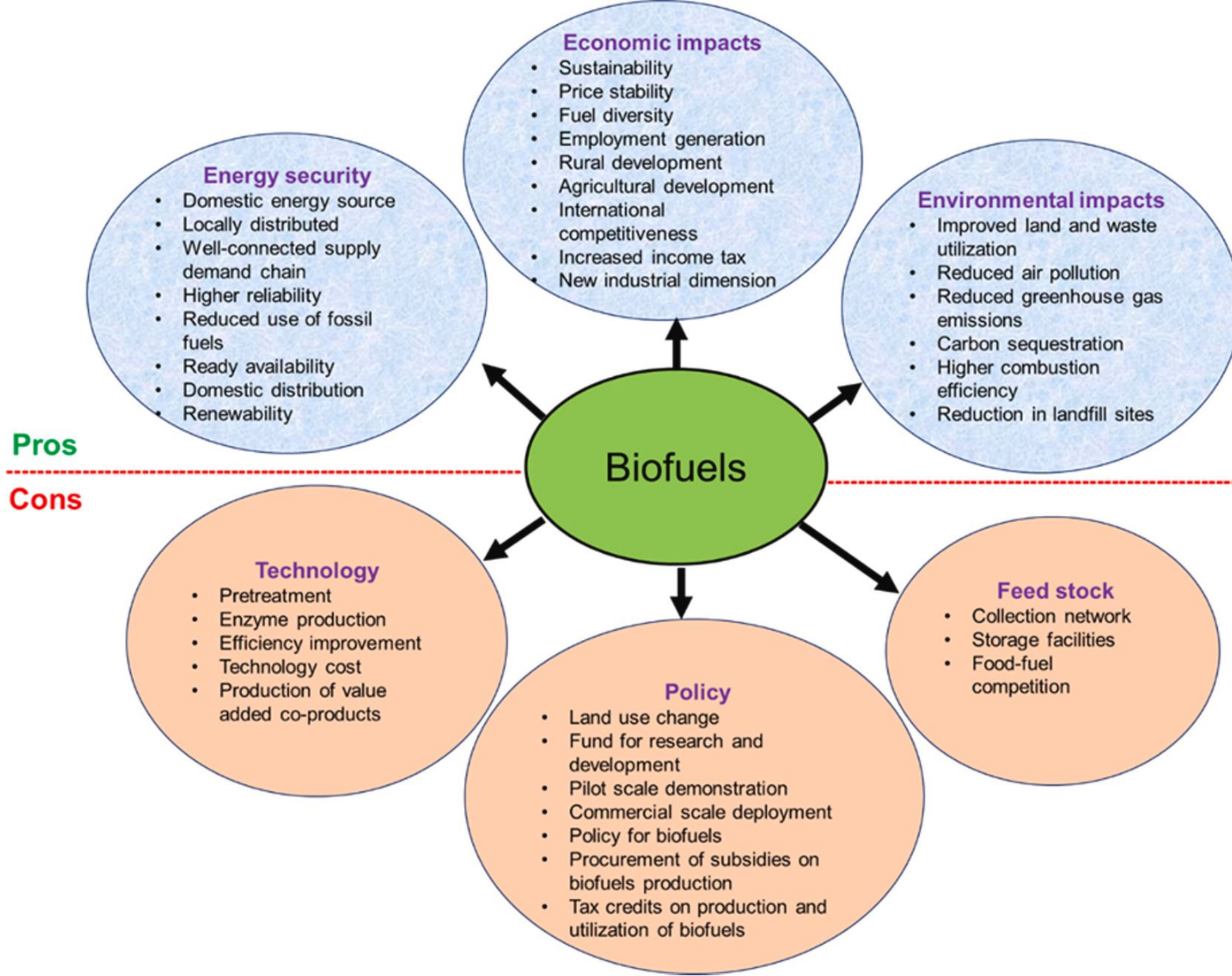


Sustainable production of bioethanol from renewable brown algae biomass



Biomass and Bioenergy, 2016, 92, 70-75

<https://www.sciencedirect.com/science/article/pii/S0961953416300903>



Answer these questions



1. Define in your own words and give examples of the following words:

- a. Biomass:
- b. Biofuel:
- c. Renewable:
- d. 1st generation biofuels:
- e. 2nd generation biofuels:
- f. 3rd generation biofuels:

2. Identify whether the following sentences are advantages or disadvantages of the different generations of biofuels, and what generation of biofuel(s) it is describing.

- 1) Burns cleaner than fossil fuels
- 2) Can use wastewater and can be grown in the ocean
- 3) Technology still needs improvement to make it competitive with fossil fuels
- 4) Biomass source are food crops such as corn, potatoes and sugar beet
- 5) Produces ethanol, which has a much lower energy density than gasoline
- 6) Competes with food crops over clean water and arable land
- 7) Biomass crops can be grown in land unsuitable for food crops
- 8) Is more energy dense per acre of harvest than the other two generations